

73) $f(x, y, z) = x^2 + y^2 - 2z^2$

Cosm

cos(2x)

Sol:-

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

$$\Rightarrow \frac{\partial}{\partial x} (x^2 + y^2 - 2z^2) + \frac{\partial}{\partial y} (x^2 + y^2 - 2z^2) + \frac{\partial}{\partial z} (x^2 + y^2 - 2z^2) = 0$$

$$\frac{\partial}{\partial x} 2x + \frac{\partial}{\partial y} 2y + \frac{\partial}{\partial z} (-4z) = 0$$

$$\frac{\partial}{\partial x} 2 + \frac{\partial}{\partial y} 2 + \frac{\partial}{\partial z} -4 = 0$$

$$2 + 2 - 4 = 0$$

$$4 - 4 = 0$$

$$0 = 0$$

74) $f(x, y, z) = 2z^3 - 3(x^2 + y^2)z$

Sol:-

$$\frac{\partial}{\partial x} 2z^3 - 3(x^2 + y^2)z \Rightarrow 2z^3 - 3x^2z - 3y^2z$$

$$\frac{\partial}{\partial x} 2z^3 - 3x^2z - 3y^2z + \frac{\partial}{\partial y} 2z^3 - 3x^2z - 3y^2z + \frac{\partial}{\partial z} 2z^3 - 3x^2z - 3y^2z$$

$$\frac{\partial}{\partial x} 0 - 6xz - 0 + \frac{\partial}{\partial y} 0 - 6yz - 0 + \frac{\partial}{\partial z} 6z^2 - 6x^2 - 6y^2 = 0$$

$$\frac{\partial}{\partial x} -6z + \frac{\partial}{\partial y} -6z + \frac{\partial}{\partial z} 6z^2 - 6x^2 - 6y^2 = 0$$

$$-6z - 6z + 12z = 0$$

$$-12z + 12z = 0$$

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75) $f(x, y) = e^{-2y} \cos 2x$

Sol:-

$$\frac{\partial}{\partial x} (e^{-2y} \cos 2x) + \frac{\partial}{\partial y} e^{-2y} \cos 2x = 0$$

~~$$\frac{\partial}{\partial x} e^{-2y} \cos 2x + \frac{\partial}{\partial y} e^{-2y} \cos 2x = 0$$~~

$$\frac{\partial}{\partial x} e^{-2y} \cdot \sin 2x \cdot 2 + \frac{\partial}{\partial y} e^{-2y} (-2) \cdot \cos 2x$$

$$\frac{\partial}{\partial x} -2 e^{-2y} \sin 2x + \frac{\partial}{\partial y} -2 e^{-2y} \cdot \cos 2x$$

$$\frac{\partial}{\partial x} -2 e^{-2y} \cos 2x \cdot 2 + \frac{\partial}{\partial y} -2 e^{-2y} \times -2 \cdot \cos 2x$$

$$\frac{\partial}{\partial x} -4 e^{-2y} \cos 2x + \frac{\partial}{\partial y} 4 e^{-2y} \cdot \cos 2x$$

~~$$-4 e^{-2y} \cos 2x + 4 e^{-2y} \cos 2x = 0$$~~

$$0 = 0$$



$$76) f(x, y) = \ln \sqrt{x^2 + y^2} \Rightarrow f(x, y) = \frac{1}{(x^2 + y^2)^{1/2}}$$

Sol:-

$$\Rightarrow \frac{1}{(x^2 + y^2)^{1/2}} \frac{\partial}{\partial x} 2x + \frac{\partial}{\partial y} \frac{1}{(x^2 + y^2)^{1/2}} 2y$$

$$\frac{\partial}{\partial x} \frac{2x}{(x^2 + y^2)^{1/2}} + \frac{\partial}{\partial y} \frac{2y}{(x^2 + y^2)^{1/2}}$$

$$\frac{\partial}{\partial x^2} \frac{2x}{(x^2 + y^2)^{1/2}} \cdot 2x + \frac{\partial}{\partial y^2} \frac{2y}{(x^2 + y^2)^{1/2}} \cdot 2y$$

$$\frac{\partial}{\partial x^2} \frac{4x^2}{(x^2 + y^2)} + \frac{\partial}{\partial y^2} \frac{4y^2}{(x^2 + y^2)}$$



77) $f(m, y) = 3m + 2y - 4$

Sol:-

$f(m, y) =$

$$\frac{\partial}{\partial m} 3 + 0 - 0 + \frac{\partial}{\partial y} 0 + 2 - 0$$

$$\frac{\partial}{\partial m} 0 + \frac{\partial}{\partial y} 0$$

$$0 + 0 = 0$$

78) $f(m, y) = \tan^{-1} \frac{m}{y}$

Sol:-

$$\frac{\partial}{\partial m} \frac{1}{1 + \left(\frac{m}{y}\right)^2} \left(\frac{m}{y}\right) \Rightarrow \frac{1}{y} \frac{1}{1 + \left(\frac{m}{y}\right)^2} \Rightarrow \frac{1}{y^2 + m^2} \cdot \frac{1}{y}$$

$$\frac{y^2}{y^2 + m^2} \cdot \frac{1}{y} \Rightarrow \frac{y}{y^2 + m^2} \Rightarrow \text{Quotient Rule}$$

$$\frac{\partial}{\partial m^2} \frac{0 \times (y^2 + m^2) - y \cdot (2m)}{(y^2 + m^2)^2} \Rightarrow \frac{0 - 2my}{(y^2 + m^2)^2} \Rightarrow -\frac{2my}{(y^2 + m^2)^2}$$



81-87 Wave equation

$$\frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 w}{\partial n^2}$$

$$81- w = \sin(n+ct)$$

Sol:-

$$\frac{\partial}{\partial t} \sin(n+ct)$$

$$\frac{\partial}{\partial n} \sin(n+ct)$$

$$\frac{\partial}{\partial t} \cos(n+ct) \cdot c$$

$$\frac{\partial}{\partial n} \cos(n+ct) \cdot 1$$

$$\frac{\partial}{\partial t} c \cos(n+ct)$$

$$\frac{\partial}{\partial n} \cos(n+ct)$$

$$\frac{\partial}{\partial t^2} -c \sin(n+ct) \cdot c$$

$$\frac{\partial}{\partial n^2} -\sin(n+ct) \cdot 1$$

$$\frac{\partial}{\partial t^2} -c^2 \sin(n+ct)$$

$$\frac{\partial}{\partial n^2} -\sin(n+ct)$$

$$-c^2 \sin(n+ct)$$

$$= \begin{aligned} & -\sin(n+ct) \times c^2 \\ & -c^2 \sin(n+ct) \end{aligned}$$

$$89) w = \cos(2\pi + 2ct)$$

Sol:-

$$\frac{\partial}{\partial t} \cos(2\pi + 2ct)$$

$$\frac{\partial}{\partial n} \cos(2\pi + 2ct)$$

$$\frac{\partial}{\partial t} -\sin(2\pi + 2ct) \cdot 2c$$

$$\frac{\partial}{\partial n} -\sin(2\pi + 2ct) \cdot 2$$

$$\frac{\partial}{\partial t} -2c \sin(2\pi + 2ct)$$

$$\frac{\partial}{\partial n} -2 \sin(2\pi + 2ct)$$

$$\frac{\partial}{\partial t} -2c \cos(2\pi + 2ct) \cdot 2c$$

$$\frac{\partial}{\partial n} -2 \cos(2\pi + 2ct) \cdot 2$$

$$\frac{\partial}{\partial t} -4c^2 \cos(2\pi + 2ct)$$

$$\frac{\partial}{\partial n} -4 \cos(2\pi + 2ct)$$

$$-4c^2 \cos(2\pi + 2ct) = -4 \cos(2\pi + 2ct) \times c^2$$

$$-4c^2 \cos(2\pi + 2ct) = -4c^2 \cos(2\pi + 2ct)$$

83) $w = \sin(n+ct) + \cos(2n+2ct)$

Sol:-

$$\frac{\partial}{\partial t} \sin(n+ct) + \cos(2n+2ct)$$

$$\frac{\partial}{\partial t} \cos(n+ct) \cdot c + (-\sin(2n+2ct) \cdot 2c)$$

$$\frac{\partial}{\partial t} \cos c(n+ct) - 2c \sin(2n+2ct)$$

$$\frac{\partial}{\partial t^2} -\sin c(n+ct) c - 2c \sin(2n+2ct) \cdot 2c$$

$$\frac{\partial}{\partial t^2} -\sin c^2(n+ct) - 4c^2 \sin(2n+2ct)$$

R.H.S

$$\frac{\partial}{\partial n^2} \sin(n+ct) + \cos(2n+2ct)$$

$$\frac{\partial}{\partial n} \cos(n+ct) (1) + (-\sin(2n+2ct) \cdot 2)$$

$$\frac{\partial}{\partial n} \cos(n+ct) - 2 \sin(2n+2ct)$$

$$\frac{\partial}{\partial n^2} -\sin(n+ct) \cdot (1) - 2 \cos(2n+2ct) \cdot 2$$

$$\frac{\partial}{\partial n^2} -\sin(n+ct) - 4 \cos(2n+2ct) \times c^2$$

$$-\sin c^2(n+ct) - 4c^2 \cos(2n+2ct)$$



84) $w = \ln(2m + 2ct)$

Sol:-

$$\frac{\partial}{\partial t} \frac{1}{(2m + 2ct)}$$

$$\frac{\partial}{\partial t} \frac{1}{(2m + 2ct)} \times 2c$$

$$\frac{\partial}{\partial t} \frac{2c}{2m + 2ct}$$

$$\frac{\partial}{\partial t^2}$$

85) $w = \tan(2m - 2ct)$

Sol:-

$$\frac{\partial}{\partial t} \tan(2m - 2ct)$$

$$\frac{\partial}{\partial t} \sec^2(2m - 2ct) \cdot -2c$$

$$\frac{\partial}{\partial t} -2c \sec^2(2m - 2ct)$$

$$\frac{\partial}{\partial t^2} -2c \cdot 2 \sec^2(2m - 2ct) \tan(2m - 2ct) \cdot -2c$$

$$\frac{\partial}{\partial t^2} -4c^2 \cdot 2 \sec^2(2m - 2ct) \tan(2m - 2ct)$$



R.H.S

$$\frac{\partial}{\partial n^2} \tan(2n - 2ct)$$

$$\frac{\partial}{\partial n} \sec^2(2n - 2ct) \cdot 2$$

$$\frac{\partial}{\partial n} 2 \sec^2(2n - 2ct)$$

$$\frac{\partial}{\partial n^2} 2 (2 \sec^2(2n - 2ct)) \tan(2n - 2ct) \times 2$$

$$\frac{\partial}{\partial n^2} 4 \sec^2(2n - 2ct) \tan(2n - 2ct) \times 2$$

$$\frac{\partial}{\partial n^2} 8 \sec^2(2n - 2ct) \tan(2n - 2ct) \times c^2$$

$$\frac{\partial}{\partial n^2} 8c^2 \sec^2(2n - 2ct) \tan(2n - 2ct)$$



$$86) w = 5 \cos(3m + 3ct) + e^{n+ct}$$

Sol:-

$$\frac{\partial}{\partial t} 5 \cos(3m + 3ct) + e^{n+ct}$$

$$\frac{\partial}{\partial t} 5(-\sin(3m + 3ct) \cdot 3c + e^{n+ct} \cdot c)$$

$$\frac{\partial}{\partial t} -15c \sin(3m + 3ct) + ce^{n+ct}$$

$$\frac{\partial}{\partial t^2} -15c \cos(3m + 3ct) \cdot 3c + ce^{n+ct} \cdot c$$

$$\frac{\partial}{\partial t^2} -45c^2 \cos(3m + 3ct) + c^2 e^{n+ct}$$

with Respect to m

$$\frac{\partial}{\partial m} 5 \cos(3m + 3ct) + e^{n+ct}$$

$$\frac{\partial}{\partial m} 5(-\sin(3m + 3ct) \cdot 3 + e^{n+ct} \cdot 1)$$

$$\frac{\partial}{\partial m} -15 \sin(3m + 3ct) + e^{n+ct}$$

$$\frac{\partial}{\partial m^2} -15 \cos(3m + 3ct) \cdot 3 + e^{n+ct} \cdot 1$$

$$\frac{\partial}{\partial m^2} -45 \cos(3m + 3ct) + e^{n+ct} \times c^2$$

$$\frac{\partial}{\partial m^2} -45c^2 \cos(3m + 3ct) + c^2 e^{n+ct}$$

